Exercise Sheet 28

Let $\mathcal{R}(K)$ be the representative-function algebra and $U = U(\mathfrak{k}_{\mathbb{C}})$. Use the coefficient pairing

$$\langle u, f\rangle = (R_u f)(e),$$

where $R_X f(g) = \partial_t|_0 f(g\exp(tX))$. The coproducts satisfy $\Delta_U X = X\otimes 1 + 1\otimes X$ and $\Delta_{\mathcal{R}} f(g,h) = f(gh)$. Exercises 28.1–28.5 are core; 28.6 is optional.

Exercise 28.1 (Circle pairing). For $K = \mathrm{U}(1)$ use $\exp(tX) = e^{it}$, $U = \mathbb{C}[X]$, and $\mathcal{R}(K) = \mathbb{C}[z, z^{-1}]$.

(a) Compute $\langle X^k, z^n\rangle$ and $\langle X^2, (z^2 + z^{-1})z^3\rangle$.
(b) Expand $\Delta_U(X^2)$ and use it to recompute the second answer by the Hopf pairing identity.
(c) For a Laurent polynomial $f = \sum_{n\in S} a_n z^n$ with fixed, known finite support $S \subset \mathbb{Z}$, show that its values $\langle X^k, f\rangle$, $0 \le k < |S|$, determine all a_n.

Exercise 28.2 (Matrix order). In $\mathfrak{sl}(2,\mathbb{C})$ let

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

For the defining representation of SU(2), extended complex-linearly at the algebra level, write $D_{ij}(g) = g_{ij}$.

(a) Compute $\langle EF, D_{11}\rangle$, $\langle FE, D_{11}\rangle$, and their difference. Compare with $[E,F] = H$.
(b) Using $\Delta D_{ij} = \sum_k D_{ik}\otimes D_{kj}$, verify the first answer by the Hopf pairing identity.
(c) For real $X \in \mathfrak{su}(2)$, show that $\langle X, SD_{ij}\rangle = -\langle X, D_{ij}\rangle$.

Exercise 28.3 (Finite dual and periodicity). For $U = \mathbb{C}[X]$, consider the two-dimensional module in which X acts by

$$A = \begin{pmatrix} i & 1 \\ 0 & i \end{pmatrix}.$$

(a) Compute A^k and show that the functional $L(p) = (p(A))_{12}$ belongs to U°.
(b) Compute $e^{2\pi A}$. Does this module integrate to the circle with $\exp(2\pi X) = e$?
(c) Why does belonging to the finite dual fail to impose the circle's global period?

Exercise 28.4 (A component invisible to derivatives). Let $K = \mathrm{O}(2)$ and $f(g) = (1 - \det g)/2$.

(a) Show that $f \in \mathcal{R}(K)$ and $f \neq 0$.
(b) Show that $\langle u, f\rangle = 0$ for every $u \in U(\mathfrak{so}(2)_{\mathbb{C}})$, including $u = 1$.
(c) Why does this not contradict the injectivity proposition for connected groups?

Exercise 28.5 (Reconstructing the centre). Let V_m be the irreducible $\mathrm{SU}(2)$ representation of highest weight m, with $\Pi_m(-I) = (-1)^m I$.

(a) Show that the family $\theta_{V_m} = (-1)^m I$, extended to direct sums, is a unitary tensor automorphism of the forgetful functor. Use the Clebsch–Gordan decomposition.

(b) What family remains after restricting to the even-weight category of $\mathrm{SO}(3)$?

(c) Explain how the defining invariant alternating tensor forces a complex tensor automorphism on $\mathbb{C}^2$ to have determinant one, and why unitarity then gives $\mathrm{SU}(2)$.

Exercise 28.6 (Optional extension: why the tensor flip works classically). Let A be a Hopf algebra with coproduct Δ, and let V, W be left modules. Their tensor product action is defined by $a(v \otimes w) = \sum a_{(1)}v \otimes a_{(2)}w$.

(a) Show that if Δ is cocommutative, then the ordinary flip $\tau(v \otimes w) = w \otimes v$ is an A-intertwiner.

(b) For $A = U(\mathfrak{k}_{\mathbb{C}})$, verify this directly on a primitive generator.

(c) If the coproduct is not cocommutative, does this calculation supply a replacement intertwiner automatically?